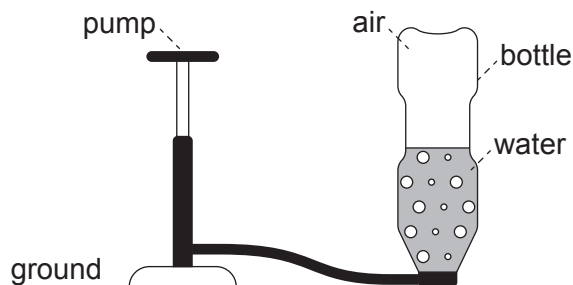


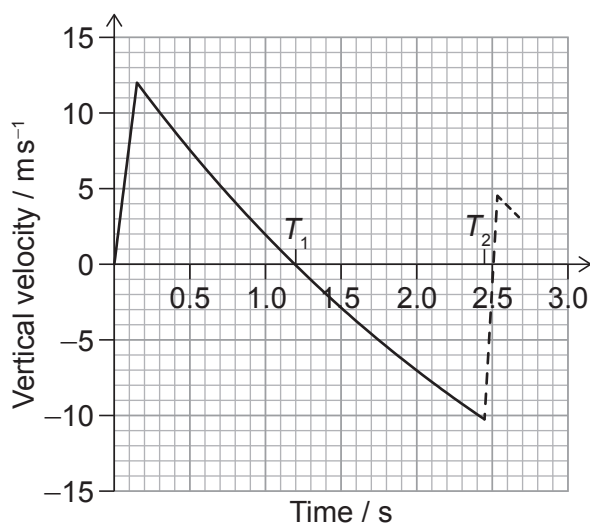
11. A toy rocket is made from a plastic bottle that contains some water.

Air is pumped into the vertical bottle until the pressure inside forces water and air out of the bottle. The bottle then travels vertically upwards.



The air–water mixture is called the propellant.

The variation with time of the vertical velocity of the bottle is shown.



The bottle reaches its highest point at time T_1 on the graph and returns to the ground at time T_2 . The bottle then bounces. The motion of the bottle after the bounce is shown as a dashed line.

- (a) Estimate, using the graph, the maximum height of the bottle.

[3]

.....

.....

.....

.....

.....

.....

(This question continues on the following page)

(Question 11 continued)

- (b) Estimate the acceleration of the bottle when it is at its maximum height. [2]

.....

.....

.....

.....

- (c) The bottle bounces when it returns to the ground.

- (i) Calculate the fraction of the kinetic energy of the bottle that remains after the bounce. [2]

.....

.....

.....

.....

- (ii) The mass of the bottle is 27 g and it is in contact with the ground for 85 ms.

Determine the average force exerted by the ground on the bottle. Give your answer to an appropriate number of significant figures. [3]

.....

.....

.....

.....

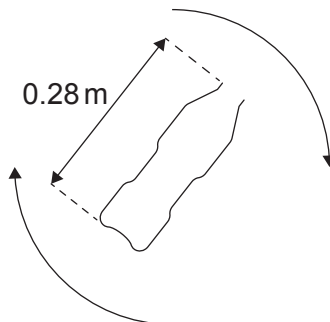
.....

.....

(This question continues on the following page)

(Question 11 continued)

- (d) After a second bounce, the bottle rotates about its centre of mass. The bottle rotates at 0.35 revolutions per second.



The centre of mass of the bottle is halfway between the base and the top of the bottle. Assume that the velocity of the centre of mass is zero.

Calculate the linear speed of the top of the bottle.

[3]

.....

.....

.....

.....

.....

.....

- (e) The maximum height reached by the bottle is greater with an air–water mixture than with only high-pressure air in the bottle.

Assume that the speed at which the propellant leaves the bottle is the same in both cases.

Explain why the bottle reaches a greater maximum height with an air–water mixture.

[2]

.....

.....

.....

.....